

Empirical Software Engineering: A gentle introduction

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**Software
Technologies
Laboratory**



This is an interactive lecture!
I will ask you questions, and
you should do the same!

Qualitative Research: The Why

...by analyzing source code repositories we will never get the complete picture of software development...

Qualitative Research: The Why

We can **only access software-centric artefacts...**

...how can we understand **developer morale?**

...how can we retrieve the **rationale of an architectural decisions?**

...how can we identify the **motivation behind deprecating a software product?**

Qualitative Research: The Why

The easiest answer:
*Ask directly who
developed the system*

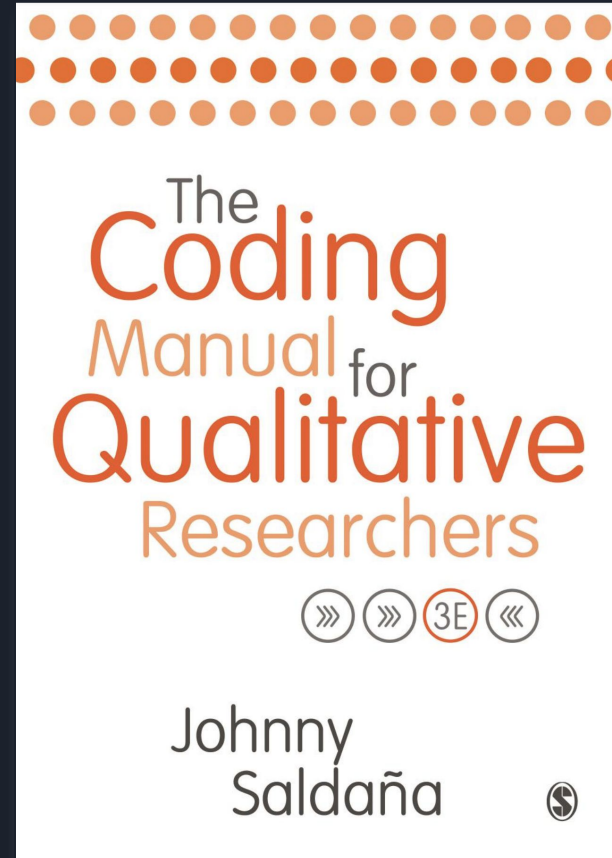
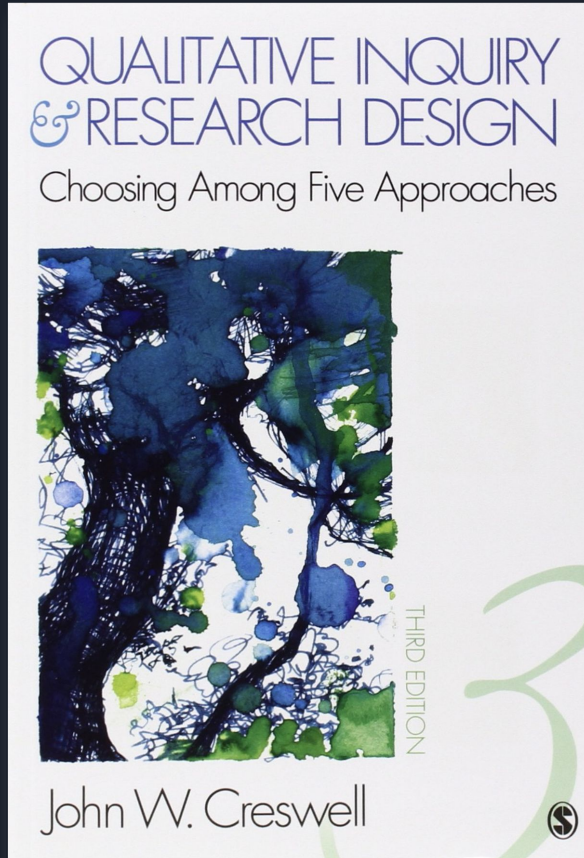


Qualitative Research: The Why

- **Not all information** is stored **in code bases**
- **Software development information** often stored in the **mind of developers**
- **Many software-related processes** are **not kept track of**
- **Qualitative information** is **not “hardcoded”**, but it is a big portion of software development
- **Quantitative** and **qualitative** data can be seen as **complementary**: one is needed to better understand and support the other

Can you give an example of
software development
process which is not
written down?

Qualitative Research: The how



Qualitative Research: The how

- Usually **does not** start from an **intuition**  
- Starts from considering a context and observing a natural phenomenon **“in the wild”**
- Mostly follows **inductive reasoning**
- **Interpret data** as it appears, **review findings**, search for **patterns** and **new directions**
- **Subjective limitations** of researchers are **mitigated with different techniques**, e.g.,
 - “Blank state” (limited knowledge of the field)
 - Bracketing (acknowledging knowledge beforehand)
 - Post-modernist stance

Qualitative Research: The how

Basic intuition of qualitative research:

1. Subject **select a specific context** and in most cases a **phenomenon**
 - a. Example context: *Development teams within a company*
 - b. Example phenomenon: *Communication between employees*
2. A **qualitative research method is selected**
 - a. Example: Focus groups
3. Relevant **participants are identified**
4. Participants are **interviewed** / given a questionnaire / ...
5. **Data is analyzed**
6. **Follow up questions** are posed (if needed)
7. Repeat steps 3-6
8. **Draw conclusions**

In your opinion, what should be considered when selecting a participant?

Participant selection

- Participant selection is a **vital part** of qualitative research
- Participants should have **knowledge of**:
 - The **context**
 - The **phenomenon**
- Strive for **heterogeneity** to better **generalize** the results
 - Usual characteristics are:
 - Age
 - Experience
 - Role
 - Company operating sector
 - Company size
 - ...

Participant selection techniques

- **Simple random sampling:** Selection is completely random from population
- **Stratified sampling:** First **divide population in groups** by some characteristics, then other sampling technique (e.g., simple random sampling)
- **Convenience sampling:** participants are selected based on availability and willingness
- **Judgement Sampling:** Researchers select participants according to their needs (e.g., *who should I talk to next?*)
- **Snowball sampling:** Participants are asked to nominate further participants
- **Each sampling techniques** has its own **pros and cons**
- **...can you name a few?**

What type of sampling is this?

Theoretical Sampling. New data is collected iteratively by purposely identifying current gaps and/or unsaturated categories of the theory. Theoretical sampling guides the selection of new data sources (e.g., participants), and the data to be collected (e.g., by generating iteratively interview questions).

Participants interviews

- Often used to **gather user feedback**
- In bigger companies, also used to **improve DevOps, develop internal tools**, etc.
- Can be **structured (close-ended)** or **unstructured (open-ended)**
- Usually last **30 to 60 minutes**
- **Interviewer and interviewee** should ideally not have **any ties (often outsourced)**
- **“Confessional”** attitude
- **Quiet** and **private environment**
- Interviews are **recorded, notes are taken** (discouraged), or both
- Depending on the method, **follow-up questions** can be asked

Structured questions: Likert scales

Response Set	1	2	3	4	5
Frequency	Never	Rarely	Sometimes	Often	Always
Quality	Very poor	Poor	Fair	Good	Excellent
Intensity	None	Very mild	Mild	Moderate	Severe
Agreement	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
Approval	Strongly disapprove	Disapprove	Neutral	Approve	Strongly approve
Awareness	Not at all aware	Slightly aware	Moderately aware	Very aware	Extremely aware
Importance	Not at all important	Slightly important	Moderately important	Very important	Extremely important
Familiarity	Not at all familiar	Slightly familiar	Moderately familiar	Very familiar	Extremely familiar
Satisfaction	Not at all satisfied	Slightly satisfied	Moderately satisfied	Very satisfied	Completely satisfied
Performance	Far below standards	Below standards	Meets standards	Above standards	Far above standards

Participants interviews: Rank places from worst to best

1. **Café:** Cozy café downtown. Closed small environment. People around are chatting, and seem to mind their own business
2. **Company office desk:** Open space office. Colleagues are sitting next by. Management has offices in another wing of the building.
3. **Company meeting room:** Meeting room of the company. The room has glass walls and is sound-insulated.
4. **Online:** Online remote setting. Interviewer and interviewee are sitting in their respective homes. Webcameras are on.
5. **Busy downtown.** Public bench on a crowded busy street downtown.

Participants interviews: Interviewer sensibility

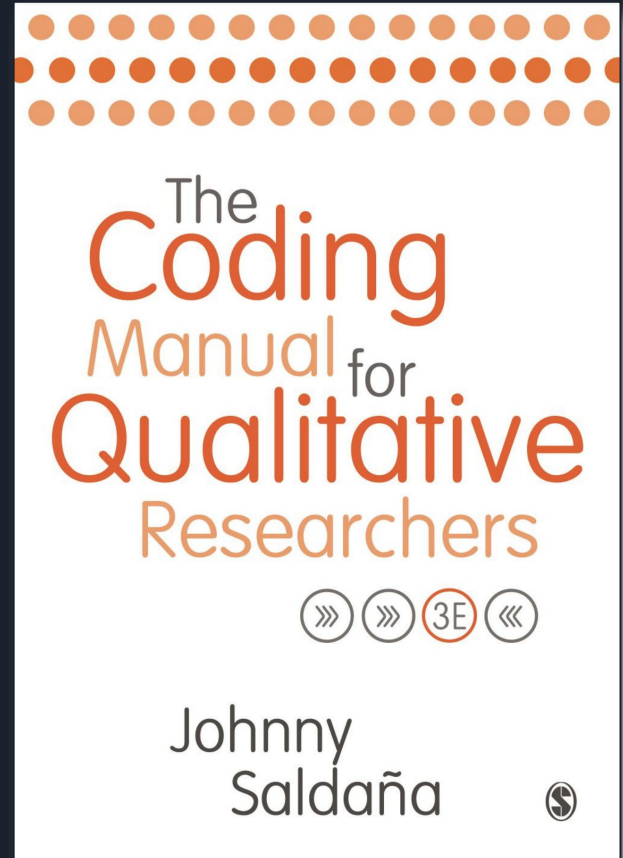
- Interviewer always needs to pay **extreme attention** to the **state of the interviewee**
- Always **steer answers/discussion** towards the **topic of interest**
- **Subtle hints** can point to **interesting directions, topics to avoid, etc.**
- Examples
 - Body language
 - Stutters
 - Pauses
 - Sounds
 - Eye contact
 - ...
- In certain settings, ensure balance between participants
- Always be **kind, polite**, and give **subtle feedback / guidance**
- **Intervene / interrupt** only when **strictly necessary**

Data analysis: Transcription

- **Taking notes** during interviews can be very efficient, but
 - Distracts interviewer
 - Interviewee might feel disregarded
 - Subtle nuances might be lost
- **Record and transcribe** can be very effective, but
 - Could be very time consuming
 - Might create vast amount of useless information
 - Interview “noise” can pollute the data
- **Automated transcription** far from perfect
 - Google Cloud API, Youtube transcriber, specialized tools, ...
 - Often misinterprets words
 - Does not account for semantics (e.g., repeated portions of sentences, broken up words, sounds, ...)
- **Denaturalism transcription approach**: grammar is corrected, interview noise (e.g., stutters) is removed and nonstandard accents (i.e., non-majority) are standardized

Data analysis: Coding

- Intuitively, **label sentences** or **paragraphs** with a **keywords** explaining their **semantic meaning**
- **Example:** *“We can use this complex patterns and more expressive data structure, and we do not consider the impact on the guy who would work on it after us”*
 - **Keywords?**

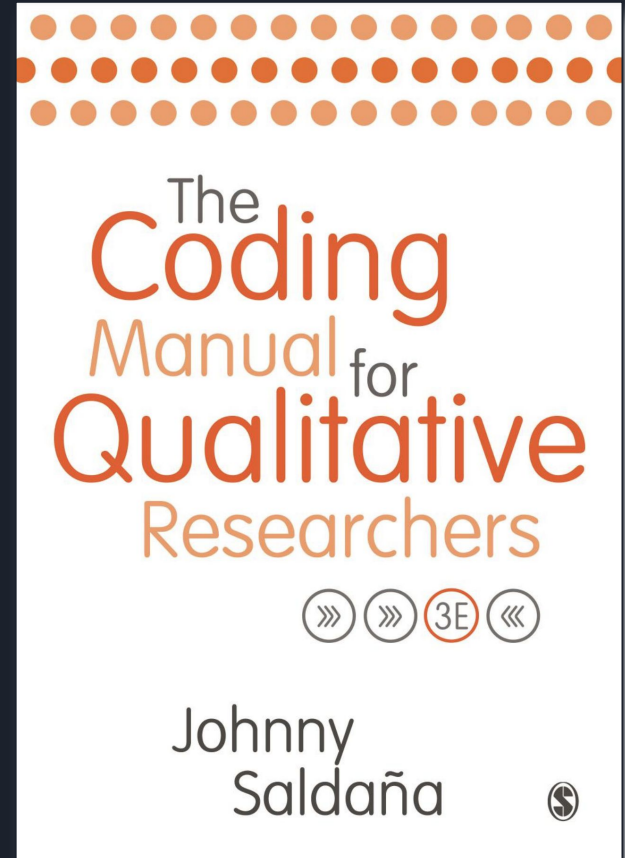


Data analysis: Let's build on it

so we started spinning off one or two teams, start to tackle the problem systematically. We wanted to cut the thing into multiple layers, and be able to work on them separately	bite_the_bullet, dedicated_team
making big disruptive changes, while keep the business moving	plane
But there were too many problems, we were fearful, and we were trying to keep the business afloat.	fear
And now, the architecting was given to a separate team I was not involved with, and due to insufficient resources in architecting, it did not come together, there was no critical mass of time and people to get it done. The re-write was never done. We lost the opportunity of be the next newest thing.	insufficient_resources
[And what was the tipping point, when debt became too much?] Scale, scale. The requirements became much higher, from 1M records it became billions, or at least hundreds of millions. And our technology could not keep up with that growth. We had to move from single to multi-threaded, and that implies a complete rewrite.	scale

Data analysis: Types of coding

- **Open coding:** segmenting data into meaningful expressions and describing them in single word to short sequence of words
- **Axial coding:** Describes a category's properties and explores how the categories and subcategories relate to each other
- **Selecting coding:** selecting one or more core categories, systematically relating them to other categories, validating those relationships, and filling in categories that need further refinement and development
- **Provisional coding:** Begins with a "start list" of researcher-generated codes
- **In-vivo coding:** Uses words or short phrases from the participant's own language in the data record as codes
- **Emotion Coding:** Labels the emotions recalled and/or experienced by the participant
- ...



Data analysis: Let's code something together!

The decision should have been that some work should have been done, the engine was a very unconventional, and worked great, but it was hard to convince the leadership that we should update things, there was no time, and so much commercial interest and opportunities.

I don't think it's very broadly understood. I brought it up in group meetings and one-on-one with certain people, but I can't feel confident to say that they like, absorbed it totally, and hold it in the same kind of severity that I do.

the C++ product reached its zenith, its goal, and it should just have been maintained without adding new functionalities, hard to communicate that to customers, because customers demand "why don't you add more features to it?", but customers don't know that adding more features to it takes longer, is harder, causes more problems etcetera in an old stack.

Data analysis: Let's code something together!

The decision should have been that some work should have been done, the engine was a very unconventional, and worked great, but it was hard to convince the leadership that we should update things, there was no time, and so much commercial interest and opportunities.

management

communication_problem

people

I don't think it's very broadly understood. I brought it up in group meetings and one-on-one with certain people, but I can't feel confident to say that they like, absorbed it totally, and hold it in the same kind of severity that I do.

lack_of_awareness,
advocate

communication_problem,
awareness

people

the C++ product reached its zenith, its goal, and it should just have been maintained without adding new functionalities, hard to communicate that to customers, because customers demand "why don't you add more features to it?", but customers don't know that adding more features to it takes longer, is harder, causes more problem etcetera in an old stack.

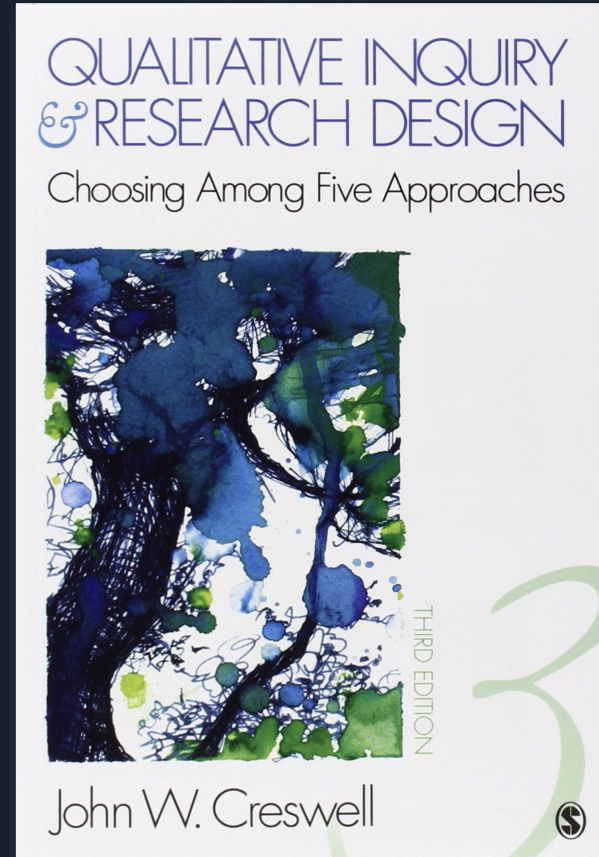
customer_perception

communication_problem

people

Types of qualitative research approaches

- *Focus groups (not in the book)*
- Grounded Theory
- Case Study
- Phenomenological Research
- Narrative Research
- Ethnographic Research



Focus groups

- **Group of people** with **certain characteristics** who generate **narrative data** in a focused discussion
- **Group dynamics and interactions are crucial** to widen range of responses, **common experiences** with different experiences and views
- **Each member shares their experiences** and compares it to the ones with the other ones
- **5 to 10 people per group** (fewer than 5 dynamics are flatter, more than 10 is difficult for the moderator to keep discussion flowing well)
- Usually starts with a **presentation by the moderator** setting the tone, **describing the topic** to be covered, etc.



Focus groups

- Group composition:
 - **Avoid power differentials**
 - **Balance heterogeneity and homogeneity:**
have a range of opinions, but also similarities to compare experiences
 - **Strangers vs. acquaintances:** depends on the study
 - **Expert vs. novices:** looking for experts in a given area with deep knowledge into a phenomenon or are we looking for a more naive view?
 - **Stratify groups** by salient characteristics (e.g., men / women)



Focus groups: Moderation advices

- **Introduce the role of the moderator** (steering when discussion gets off topic, can summarize answers to make sure the shared viewpoint is clear, etc.)
- Use **conversational, clear, simple language**, not convoluted scientific rigmarole (repeat sentences aloud before study group, to see if these sound fitting or off putting)
- **Open ended questions are crucial**. Should not lead in any way to a specific answer, for the respondents to lead the way and shape the direction of the discussion
- A technique is the **“think back”**, e.g. **“think back to the last time you experienced this”**
- **Avoid to ask “why”**. Even if it is the end goal, it is very sharp in tone, may cause defensiveness, and sound like there is a correct answer. Instead, focus on “how” things happened, describe their experience in detail
- **Avoid giving examples** if not strictly needed to get people going, but it risks to lead people down a particular path
- **Be aware of the participants limits** (time, attention, language, cultural issues, communications skills)

In a focus group, a senior manager is very experienced and talkative. They always take the word.

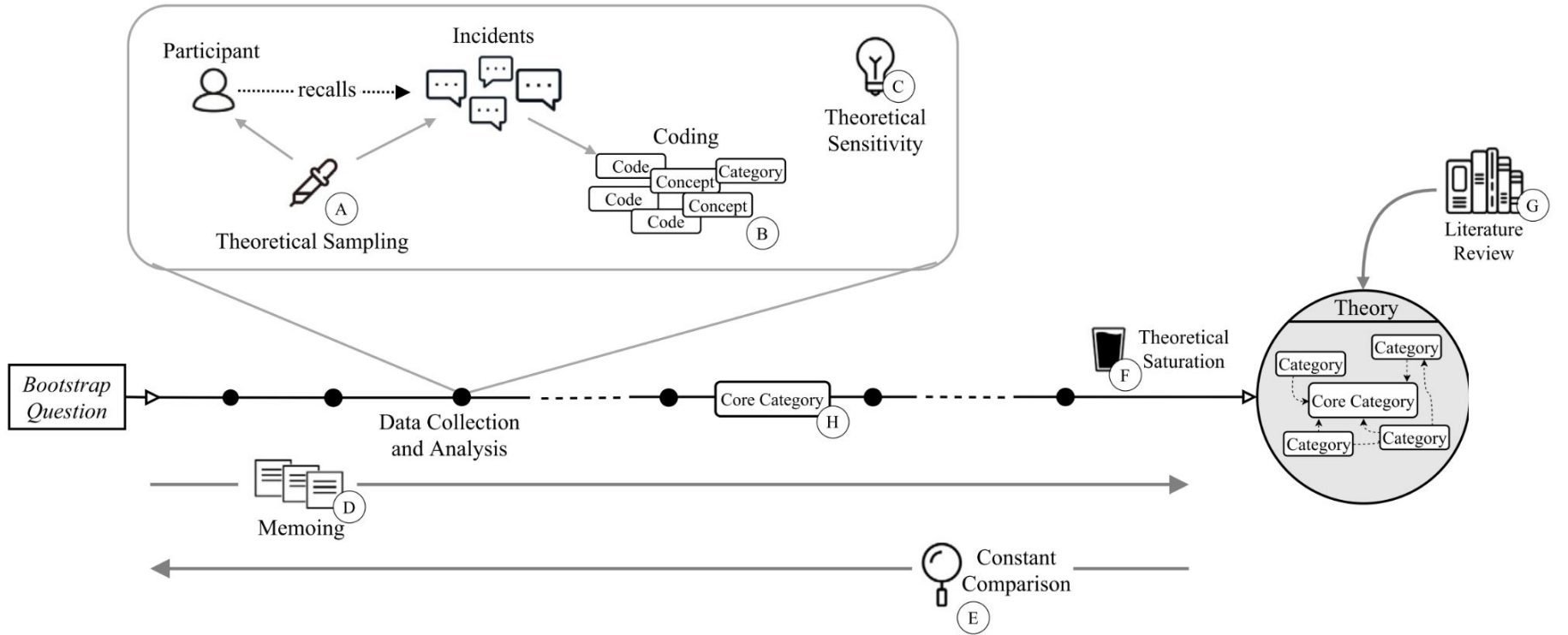
Should I intervene? Why? How?

Grounded theory

- **Construction** of **hypotheses** and **theories** by *grounding* the findings in collected and analyzed qualitative data
- Goal: answer the question **“What is going on here?”**
- **Different stances** (e.g., Glaserian, Straussian, constructivist, ...)
- Process overview:
 - **“Incidents”** (i.e., bits of gathered data related to the studied phenomenon) are **collected and analyzed** to **identify emerging concepts**
 - **Concepts are aggregated semantically** into different **categories** (the basic building blocks of the emerging theory).
 - **Categories are further developed** by gathering additional data and comparing the new incoming incidents against the old ones already categorized.
 - Process leads to the identification of **abstract categories fitting most data**



Grounded theory



Grounded theory: Core concepts

- **Theoretical Sampling.** New data is collected iteratively by purposely identifying current gaps and/or unsaturated categories of the theory
- **Theoretical Sensitivity.** Ability of the researcher guiding the theoretical sampling, conceptualization of incidents, and identification of relations between concepts
- **Memoing.** Memos are used to keep track of emerging concepts/categories, the relations between them, and potential gaps in the theory
- **Constant comparison.** Throughout the entirety of the study, all artifacts (incidents, codes, concepts, categories, and memos) are constantly compared and updated.
- **Theoretical Saturation.** Adding new data does no longer result in an update of the established theory.



Towards the end of a market research, a user tells me a new feature that's was never mentioned before should be added to my app.

What should I do? Why?

Case study

- Approach used to generate an **in-depth, multi-faceted understanding** of a **complex issue** in its **real-life context**
- **Detailed, in-depth data collection** involving **multiple sources** of information (e.g., observations, interviews, audiovisual material, and documents and reports)
- Process overview
 - Begins with the **identification of a specific case**
 - **Goal** can be either:
 - **Illustrate a unique case**
 - **Understand a specific issue**, problem, or concern
 - In-depth understanding of the case by gathering **many forms of qualitative / quantitative data**
 - Can consider **holistic analysis** (entire case) or **embedded analysis** (specific aspect of the case)
- Ends with an **in-depth description of the case**, followed potentially by **focus on key issues**, and **comparison across cases**



Phenomenology

- **Goal:** identify the **common meaning** for several individuals of their lived experiences **of a concept or a phenomenon**
- Develops a **composite description of the essence of the experience** for all of the individuals
- Process overview:
 - Define **research problem** to investigate
 - **“Bracketing”**: investigators documents and sets aside their experiences
 - **In-depth and multiple interviews** with participants
 - **What have you experienced** *in terms of the phenomenon?* **(WHAT)**
 - **What contexts or situations have** typically **influenced** or **affected your experiences of the phenomenon?** **(HOW)**
 - [optional] Clarifying open-ended questions
- Data analysis
 - **Horizonalization** (identify significant statements), **clusters of meaning, textural & structural description**, essential, invariant structure (or **essence**)



In the context of *testing AI-based systems*, make up an answer to:

What have you experienced in terms of the phenomenon? (WHAT)

What contexts or situations have typically influenced or affected your experiences of the phenomenon? (HOW)

Threats to validity in empirical research



Threats to validity

- While striving to be correct, **empirical studies do not provide formally / mathematically proven results**
- **Correctness of results** are supported by the:
 - **Selection** of appropriate **research method**
 - **Systematicity** in following the method
 - Clear and complete **documentation** of the study and the results
- **Factors** that might have **influenced the results** reported as **threats to validity in a dedicated section**
- Usually, follows the formulation:
 - **Description** of the threat
 - **How it could have influenced the results**
 - **Mitigation strategy** (or **motivation** why none was adopted)



Example of threat to validity and mitigation strategy

A prominent threat to the external validity of our study is posed by the adoption of a single experimental subject and a subset of AI algorithms. To mitigate this threat, we chose our experimental subject and independent variables to be as representative as possible. Specifically, we selected a common AI classification problem, namely text classification, selected a widely utilized peer-reviewed dataset [29], and considered a total of 6 different algorithms provided in the largely adopted Python library `scikit-learn`.

Types of threats to validity

- **Conclusion Validity:** Threats to the conclusion validity are concerned with issues that affect the **ability** to draw the **correct conclusion about relations** between the **treatment** and the **outcome** of an experiment.
- **Internal Validity:** Threats to internal validity are **influences that can affect the independent variable** with respect to causality, without the researcher's knowledge. Thus they threaten the conclusion about a possible causal relationship between treatment and outcome.
- **Construct Validity:** Construct validity concerns **generalizing the result of the experiment to the concept or theory behind the experiment**. Some threats relate to the design of the experiment, others to social factors.
- **External Validity:** Threats to external validity are conditions that **limit our ability to generalize the results** of our experiment to industrial practice

Claes Wohlin · Per Runeson
Martin Höst · Magnus C. Ohlsson
Björn Regnell · Anders Wesslén

Experimentation in Software Engineering

 Springer

I used a single experimental object
for my study.

What type of threat is this? What
could be a potential mitigation
strategy?

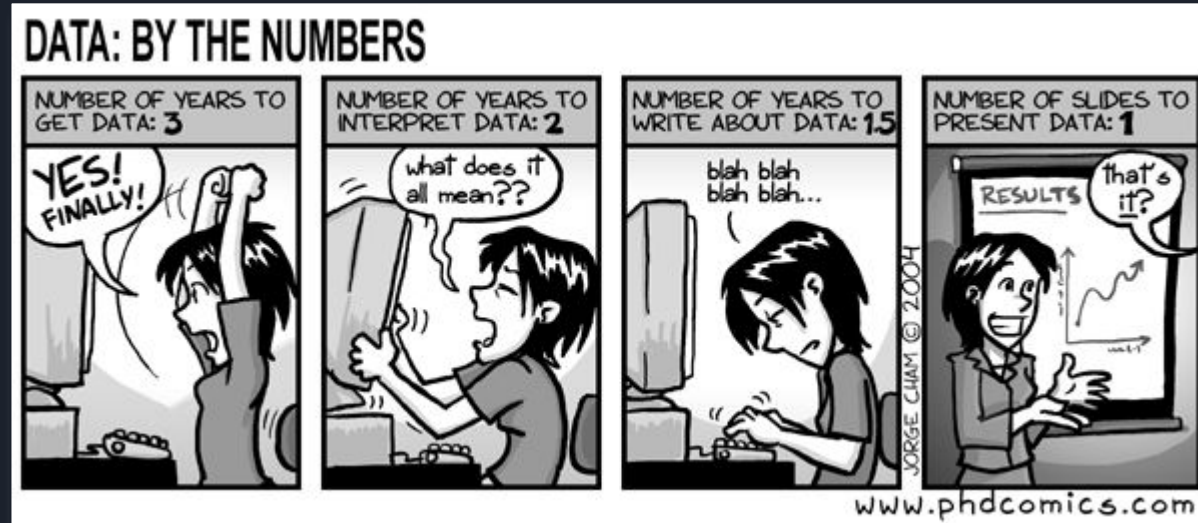
In my experiment, history affects the results, as the circumstances are not the same at different times.

What type of threat is this? What could be a potential mitigation strategy?

Documenting empirical studies: Basic structure

- (Structured) Abstract
- Introduction/Motivation
- (Related Work)
- Experiment design
- Experiment execution
- Data analysis
- Threats to validity
- Discussion
- Conclusions
- References

Documenting empirical studies (*and how your project should be written* 😊)



Documenting empirical studies: Abstracts

Abstract MadLibs!!

This paper presents a _____ method for _____
(synonym for new) (sciencey verb)
the _____. Using _____, the
(noun few people have heard of) (something you didn't invent)
_____ was measured to be _____ +/- _____
(property) (number) (number)
_____. Results show _____ agreement with
(units) (sexy adjective)
theoretical predictions and significant improvement over
previous efforts by _____, et al. The work presented
(Loser)
here has profound implications for future studies of
_____ and may one day help solve the problem of
(buzzword)
_____.
(supreme sociological concern)

Keywords: _____, _____, _____
(buzzword) (buzzword) (buzzword)

WWW.PHDCOMICS.COM

Documenting empirical studies: Abstracts

- **People judge studies by their abstracts**
 - Usually decide whether to read the whole paper based on abstract
- It's important for the abstract to **tell the whole story**
 - Background and context
 - Goals
 - Method
 - Results
 - Conclusion
- Often it is the **only part freely accessible**

The structured abstract

- **Context:** Brief explanation of the motivation for conducting the study
- **Objective:** Aim, objects, focus, perspective of the study (based on the GQM)
- **Method:** Experimental design, number and kind of objects/subjects, selection criteria, data collection and analysis procedures
- **Results:** Main findings
- **Conclusion:** Impact of the obtained results

Example of structured abstract

Asking about Technical Debt

Characteristics and Automatic Identification of Technical Debt Questions on Stack Overflow

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ABSTRACT

Background: Numerous methodologies have been used to study technical debt. Among different data sources, Q&A sites provide an opportunity to study how users reference and request support on technical debt. To date only few studies, focusing on narrow aspects, investigate technical debt through the lens of Stack Overflow.

Aims: We aim at gaining an in-depth understanding on the characteristics of technical debt questions on Stack Overflow. In addition, we assess if identification strategies based on machine learning can be used to automatically identify and classify technical debt questions.

Method: We use combination of automated and manual processes to identify technical debt questions on Stack Overflow. The final set of 415 questions is analyzed both quantitatively and qualitatively to study (i) technical debt types, (ii) question length, (iii) perceived urgency, (iv) sentiment, and (v) emerging themes. Natural language processing and machine learning techniques are used to evaluate if technical debt questions can be identified and classified automatically.

Results: Architecture debt is the most recurring debt type, followed by code and design debt. Most questions display mild urgency, with frequency of higher urgency steadily declining as urgency rises. Question length varies across debt types. Sentiment is mostly neutral. 29 recurrent themes emerge. Machine learning can be used to identify technical debt questions and binary urgency, but not debt types.

Conclusions: Different patterns emerge from the analysis of technical debt questions on Stack Overflow. The results provide further insights on the phenomenon, and support the adoption of a more comprehensive strategy to identify technical debt questions.

1 INTRODUCTION

During the past years, a plethora of different research methods and data sources have been used to study Technical Debt (TD). Among these studies, quantitative investigations relying on source code analysis were frequently used to identify and manage TD [7, 15, 21, 35, 41, 61]. Researchers also used qualitative approaches, such as surveys, focus groups, and grounded theory, to acquire knowledge from those directly experiencing TD and gain insights that could be missed by utilizing exclusively quantitative approaches [25, 32, 34, 49, 59]. Among the various data sources used, a vast pool of data which may contain information useful to further understand TD remains almost uncharted. With over 23M questions asked, 18M registered users, and 11M visits per day¹, Stack Overflow (SO) ranks top in all these metrics among the popular umbrella of Stack Exchange Question and Answer (Q&A) sites. Possibly, SO is to date largest and most popular Q&A site where users ask questions, share knowledge, and discuss topics related to computer programming. Despite its large popularity, relatively few research has investigated TD from the point of view of SO.

In this study, we aim at understanding the nature of questions regarding technical debt posted on Stack Overflow, which from now on are referred to as Technical Debt Questions (TDQs). Specifically, we aim at characterizing the TD phenomenon through the lens of SO by studying different characteristics of TDQs, such as TD types, question length, urgency, sentiments, and considered themes. In addition, since we have to rely on a keyword-based search followed by manual analysis to identify TDQs, we also investigate the potential of predictive models for automatically identifying and categorizing TDQs.

With regards to scientific progress, our study constitutes a first

The introduction and motivation section

- **Introduction**

- **Mini-version of the whole paper**

- Attacked problem
 - Proposed solution or experiment
 - Main results
 - Main contributions (how to “use the paper”)
 - Structure of the paper

- **Motivation**

- Scope of the work
 - Problem statement
 - Research objectives
 - Encourages to read the paper

The related work section

- *Why this study is necessary?*
- Complete picture about:
 - **experiments** with **similar goals**
 - **experiments** with **similar objects/subjects**
 - **papers with similar goals, but not empirical** → they lack in provided evidence
- Two main strategies:
 - **Paper-by-paper** comparison
 - **Catalogue of related papers + overall comparison**

The related work section: Further tips

- Identify a **couple of important “topics”** relevant to your research
- For each topic, write about
 - the **TRAJECTORY** of the research progress as a story and
 - the **RELATIONSHIP** of prior art and this paper
- **Avoid laundry list**
- **Don't just describe, RELATE it**
- **Be respectful**
- **Be generous**

The experiment design and execution section

- **Design: rewalk through all the experiment plan**
 - GQM
 - Hypotheses
 - objects and subjects
 - Variables
 - Measurements
 - experiment design
 - data analysis strategies (e.g., used statistical tests)
- **Execution: how the experiment plan has been put in practice**
 - preparation (eg code instrumentation)
 - data collection procedure
 - data clean up
 - any deviation from the plan
 - reference to replication package

The results and threats to validity sections

- **Results**

- Demographics
- Descriptive statistics and data exploration
- For each research question:
 - hypothesis testing
 - discussion of effect size
 - quick elaboration on obtained results

- **Threats to validity**

- External
- Internal
- Construct
- Conclusion

BONUS! 🔥 Online book on how to better visualize your results: <https://serialmentor.com/dataviz/>

The discussion and conclusion sections

- **Discussion**

- Elaborate on the obtained results
- Impact into practice (industry)
- Make the results actionable
 - e.g., how will app developer John use your results tomorrow?
 - e.g., what researcher Y learned about phenomenon X?
- Lessons learnt

- **Conclusions**

- Summary of the main contributions of the paper
- Future work
 - other experiments on different aspects
 - better ways to perform the experiment
 - ...

The replication package

- **Data**

- raw data (usually CSV)
- processed data (CSV)
 - (name it properly!)

- **Source code (commented!)**

- measurement scripts (Python)
- data processing scripts (R)
- analysis scripts (R)

- **Figures**

- Exploratory figures
- Final figures

- **Text**

- README file
- (optional) experiment notes

Example of replication package: <https://github.com/TD-SO/rep-pkg>

Replication package template: <https://github.com/S2-group/template-replication-package>

Example of tool repository: <https://github.com/STLab-UniFl/oreo-tool>

Two words on the final project: Overall structure

- **Quantitative or qualitative** inquiry on a **software engineering topic**
- Can be conducted in **teams** (max. 3 people)
- **Weekly joint “stand-ups”** to present progress, discuss impediments, ...
- Final “exam”:
 - **Structured deliverable** documenting the analysis and the results, accompanied by a “replication package”, and
 - **Hold a presentation** of the results
- Project can naturally develop in a thesis subject
- High quality deliverables could be reworked into a scientific publication



Two words on the final project: Regarding the topic of choice

- You must choose a topic that **interests you**
- The topic should have a **practical motivation** (e.g., no *"On the Evolution of the "hack" Keyword in README Files"*)
- **Read some papers before jumping into the project.**
Be creative, be curious, be smart
- If you have a loose idea, the **topics can jointly finalized with me**
- If clueless, **I can provide ideas**
- The final topic choice is yours. However I can support you most on topics related to **software quality, software architecture, green software, technical debt, testing, software education**
- **"Everything" goes:** StackOverflow, ChatGPT, chatting with your lead architect friend, call at home, etc.
However, you have to know what you are doing, and not commit plagiarism.
- Contact me immediately if "lifting" instances develop (not at the end of the project)



*Final word on the project: Above and beyond,
the sky is (not?) the limit*

You

1337 h4x0r! It worked magic!

Once my lecturer at my 2nd undergraduate year said:

"There are two types of people: normal ones and wizards. You are supposed to be the wizards" 🧙

11:46

Do you have a question or doubt? Reach out!

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Roberto Verdecchia
*Computer Scientist,
Assistant Professor at
UniFI.*

Hi there! I am **Roberto Verdecchia**, an Assistant Professor at the [STLAB](#) of the [University of Florence](#).

My research interest lies in the intersection between *Software Architecture* and *Technical Debt* (which combined can look something like [this](#)). I am also fascinated by topics related to *Software Sustainability*, *Software Testing*, and *Software Education*.

Prior to joining the [STLAB](#), I was a Postdoctoral Research Associate of the [S2Group](#) at the [Vrije Universiteit Amsterdam](#).

I hold a Double Ph.D. obtained at the [Computer Science Department](#) of the [Gran Sasso Science Institute](#) and at the [S2Group](#) of the [Vrije](#)



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SCAN ME

<https://robertoverdecchia.github.io>

Oh look! We still have heaps of time!

Let's make the best out of it!

- **Option 1**

Empirical research is hard!
Let's look at one together!

https://robertoverdecchia.github.io/papers/ICT4S_2022_1.pdf

- **Option 2**

Got all the basics of
empirical research?
Let's design a study together!